

9	(a) (induced) e.m.f. proportional to rate of change of (magnetic) flux (linkage)	M1 A1 [2]
	(b) flux linkage = BAN	
	$= \pi \times 10^{-3} \times 2.8 \times \pi \times (1.6 \times 10^{-2})^2 \times 85 = 6.0 \times 10^{-4} \text{ Wb}$	B1 [1]
	(c) e.m.f. = $\Delta N\Phi / \Delta t$	
	$= (6.0 \times 10^{-4} \times 2) / 0.30$	C1
	$= 4.0 \text{ mV}$	A1 [2]
	(d) sketch: $E = 0$ for $t = 0 \rightarrow 0.3 \text{ s}$, $0.6 \text{ s} \rightarrow 1.0 \text{ s}$, $1.6 \text{ s} \rightarrow 2.0 \text{ s}$	B1
	$E = 4 \text{ mV}$ for $t = 0.3 \text{ s} \rightarrow 0.6 \text{ s}$ (either polarity)	B1
	$E = 2 \text{ mV}$ for $t = 1.0 \text{ s} \rightarrow 1.6 \text{ s}$	B1
	with opposite polarity	B1 [4]

Page 7	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2016	9702	41